# Real Weighted-MC Atmospheric $\theta_{23}/\Delta m^2_{3l}$ Inference at IceCube DeepCore with `tpeanuts`

Fast Simulation of Neutrino Oscillations in Matter

MSc Thesis (Trabajo Fin de Máster) — Inference Companion Document

Juan Ramon Diaz Santos
diazjuan@alumni.uv.es

Supervisors: Roberto Ruiz de Austri Bazan & Michele Lucente

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This document accompanies `notebooks/inference/inference5_icecube.ipynb` and the `tpeanuts.detector.icecube`/`tpeanuts.inference` code packages: it documents, end to end, the real IceCube DeepCore 9-year "golden event" atmospheric-neutrino public data release, the event-by-event weighted-Monte-Carlo reweighting forward model implemented in `tpeanuts` (unlike every other detector in this project, which folds a continuous differential cross section), the real detector-systematics hypersurface correction, the gradient-based Poisson fit, and the resulting real $\theta_{23}/\Delta m^2_{3l}$ measurement, compared against IceCube's own published best fit – together with a dedicated audit of the fitted muon-background normalization scale against IceCube's own published value. Blue boxes (**Theoretical Foundation**) contain the physics; green boxes (**Implementation**) link that physics to actual `tpeanuts` code; orange boxes collect approximations and limitations; cyan boxes (**Result**) report a concrete numerical outcome of the notebook or of this project's own test suite. Text highlighted in yellow marks key results/equations.

# Contents


**1 Introduction to IceCube DeepCore** **1**

1.1 Purpose and how to read this document . . . . . . . . . . . . . . . . . . . . . . 1

1.2 Overview . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 1

1.3 Atmospheric neutrino production and the oscillation signal . . . . . . . . . . . 2

1.4 Event-by-event weighted Monte Carlo: why IceCube is different . . . . . . . . . 2

1.5 Data-taking period and physics goals . . . . . . . . . . . . . . . . . . . . . . . 3

**2 The Official Data Release: Tables and Publications** **4**

2.1 Publications . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 4

2.2 Data structure: event-by-event weighted Monte Carlo . . . . . . . . . . . . . . . 4

2.3 Table catalogue . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 5

**3 Detector Implementation in `tpeanuts`** **6**

3.1 Weighted Monte Carlo reweighting . . . . . . . . . . . . . . . . . . . . . . . . . 6

3.2 Real 3-flavour Earth-matter oscillation . . . . . . . . . . . . . . . . . . . . . . 7

3.3 Real atmospheric flux and detector-systematics hypersurface . . . . . . . . . . . 7

3.4 Free normalization parameters . . . . . . . . . . . . . . . . . . . . . . . . . . 8

**4 The Inference Process** **10**

4.1 Normalization calibration . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 10

4.2 Poisson fit statistic and gradient-based procedure . . . . . . . . . . . . . . . . . 10

4.3 This document's own test suite . . . . . . . . . . . . . . . . . . . . . . . . . . 11

4.4 The muon-background normalization audit . . . . . . . . . . . . . . . . . . . . 12

**5 Results** **13**

5.1 Detector-level closure check . . . . . . . . . . . . . . . . . . . . . . . . . . . . 13

5.2 Real fit . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 13

5.3 2-D likelihood slice . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 14

5.4 Real data comparison . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . 14

5.5 The muon-background normalization audit . . . . . . . . . . . . . . . . . . . . 15

5.6 Summary and known limitations . . . . . . . . . . . . . . . . . . . . . . . . . 15

Chapter 1

# Introduction to IceCube DeepCore

## 1.1 Purpose and how to read this document

This document is the third inference-side companion to the *Theoretical Foundations of* ***tpeanuts*** document: where *Inference at Daya Bay* documents a reactor antineutrino application and *Real Salt-Phase (SNO-II)* documents a solar-neutrino application, this one documents an *atmospheric*-neutrino application, built on the only detector in this project whose real public data is event-by-event weighted Monte Carlo rather than a continuous cross-section fold – fitting the atmospheric mixing angle $\theta_{23}$ and the mass-squared splitting $\Delta m^2_{3l}$ to the real public data release of the IceCube DeepCore experiment. Every number quoted here that is not explicitly attributed to a fitted `tpeanuts` result is a real, published, externally verifiable quantity, traceable through the References chapter.

The five chapters are organised as follows: this chapter introduces the experiment itself – geometry, detection principle, and data-taking period – with no reference yet to `tpeanuts`; Chapter 2 catalogues the real official public data release the code consumes; Chapter 3 is the physics-and-mathematics core, developing the weighted-Monte-Carlo reweighting formula and the detector-systematics hypersurface correction `tpeanuts.detector.icecube` implements; Chapter 4 documents the Poisson fit machinery and this document's own dedicated test suite and audit work; Chapter 5 presents, interprets, and summarises the numerical results obtained from `inference5_icecube.ipynb`, compared against IceCube's own published best fit, together with the muon-background normalization audit this document's own working session carried out.

## 1.2 Overview

IceCube is a cubic-kilometre neutrino telescope instrumenting the glacial ice at the geographic South Pole with 5160 digital optical modules (DOMs) on 86 vertical strings, detecting the Cherenkov light emitted by charged particles produced in neutrino interactions within or near the ice. **DeepCore** is a denser sub-array of 8 additional strings deployed in the clearest, deepest ice near the bottom centre of the main array, lowering IceCube's effective energy threshold from $\sim$100 GeV down to a few GeV – the sub-array this document's entire analysis is built on [1]. This document concerns the real 9-year "golden event" sample: well-reconstructed, predominantly track-like (muon-neutrino charged-current) events, selected for their high purity and good angular/energy reconstruction, used by the collaboration's own atmospheric-oscillation analysis.

**Approximations and Limitations**

**Scope of this document.** `tpeanuts` reproduces IceCube's own real event-by-event weighted-Monte-Carlo reweighting technique, a real 3-flavour Earth-matter oscillation probability, the real Honda atmospheric flux calculation, and the real per-bin detector-systematics hypersurface correction, evaluated at the collaboration's own published best-fit systematics point. It does *not* re-fit the 5 detector-systematics parameters as free nuisances (they are held fixed, a deliberate, documented scope decision – Chapter 3), nor reproduce a full atmosphere-production-height treatment (production is treated at the top of the atmosphere, a sub-percent-level simplification). Chapter 5 reports, quantitatively and without adjustment, what these simplifications do and do not explain about a real discrepancy between this document's own fitted muon-background normalization and IceCube's own published value.

## 1.3 Atmospheric neutrino production and the oscillation signal

**Theoretical Foundation**

Atmospheric neutrinos are produced by cosmic-ray-induced particle showers in the upper atmosphere: charged pions and kaons decay into muons and muon neutrinos, and the muons themselves subsequently decay, producing further muon and electron neutrinos – a well-understood, precisely calculated flux [5] spanning many decades in energy and the full range of zenith angles. Neutrinos arriving from below the horizon (upgoing) have crossed some or all of the Earth's diameter since production, at baselines ranging from a few hundred kilometres (near horizontal) to ∼12,700 km (straight through the core) – a natural, continuously-varying baseline that, combined with the atmospheric flux's own wide energy range, makes the upgoing $\nu_\mu$ disappearance pattern a direct, high-statistics probe of $\theta_{23}$ and $\Delta m^2_{3l}$ (the same oscillation minimum first resolved by Super-Kamiokande's own 1998 atmospheric-neutrino discovery, refined here with IceCube DeepCore's own real GeV-scale event sample and a real Earth-matter treatment).

**Approximations and Limitations**

Downgoing (production directly overhead, negligible propagation baseline) and near-horizontal events carry little oscillation information; this document's real analysis binning (Chapter 2) accordingly spans $\cos\theta_z \in [-1, 0.1]$ (upgoing to just past horizontal) only, matching the real public release's own binning exactly.

## 1.4 Event-by-event weighted Monte Carlo: why IceCube is different

**Theoretical Foundation**

Every other detector in this project (Daya Bay, Borexino, SNO) provides either a continuous differential flux/cross section or an officially extracted pseudo-observable, folded through a detector response using `tpeanuts.detector.common.event_rate`. IceCube's own public data release provides neither: instead, it ships ∼400,000 individually simulated Monte Carlo events, each carrying its own *true* (simulated) energy and direction, its *reconstructed* analysis-bin location, and a real physical weight (units GeV cm$^2$ sr) that must be multiplied by the atmospheric flux and a livetime to become a physical event count – the standard "weighted Monte Carlo" technique used throughout atmospheric-neutrino oscillation analyses, and the reason `tpeanuts.detector.icecube` reweights real simulated events directly (Chapter 3) rather than reusing the continuous-fold machinery every other detector package in this project shares.

## 1.5 Data-taking period and physics goals

**Theoretical Foundation**

The real public data release used throughout this document spans 9 years of DeepCore operation, and reports IceCube's own real published best fit (Table III/IV of [1], normal ordering, for later comparison in Chapter 5):

$$\sin^2\theta_{23} = 0.51, \qquad \Delta m^2_{32} = 2.41^{+0.07}_{-0.07}\times 10^{-3}\,\mathrm{eV}^2, \qquad \text{atmospheric-muon norm. scale} = 1.39, \tag{1.1}$$

with $\theta_{23}$ measured via the atmospheric $\nu_\mu$ disappearance minimum's own energy/zenith-angle position, and the muon-background normalization scale a real, published nuisance-parameter result of the collaboration's own joint fit (not an oscillation parameter, but a detector/background quantity this document's own Chapter 5 audits directly against its own fitted value).

Chapter 2

# The Official Data Release: Tables and Publications

## 2.1 Publications

**Key References**

- **Primary source.** R. Abbasi et al. (IceCube Collaboration), *Measurement of atmospheric neutrino mixing with improved IceCube DeepCore calibration and data processing*, Phys. Rev. D **108**, 012014 (2023), arXiv:2304.12236 [1]. Its official public data release (Harvard Dataverse, DOI 10.7910/DVN/B4RITM) is the sole source of every real table this project's `data/detector/icecube/` tree consumes.
- **Atmospheric flux.** M. Honda, T. Kajita, K. Kasahara and S. Midorikawa, *Atmospheric neutrino flux calculation using the NRLMSISE-00 atmospheric model*, Phys. Rev. D **92**, 023004 (2015), arXiv:1502.03916 [5]. Source of the real 3-dimensional atmospheric flux table, already cached by this project's own `tpeanuts.source.atmosphere` package (not fetched anew for this document).
- **External comparison: NuFIT.** I. Esteban et al., *The fate of hints: updated global analysis of three-flavor neutrino oscillations*, JHEP **12**, 216 (2024), arXiv:2410.05380 [4] – base analysis publication for NuFIT 6.1 (2025), `www.nu-fit.org`. Source of the fixed $(\theta_{12}, \theta_{13}, \Delta m^2_{21}, \delta_{CP})$ values and the starting $(\theta_{23}, \Delta m^2_{3l})$ point used throughout Chapters 3 and 5.

## 2.2 Data structure: event-by-event weighted Monte Carlo

**Theoretical Foundation**

Data provenance: `notebooks/external/icecube/IceCube1_generator.ipynb` fetches and validates the release verbatim into `data/detector/icecube/raw/`. Four physically distinct interaction channels are shipped as separate real Monte Carlo event tables – neutral-current (`nc`), electron-neutrino charged-current (`nue_cc`), muon-neutrino charged-current (`numu_cc`), and tau-neutrino charged-current (`nutau_cc`) – each event carrying its own true energy, true $\cos\theta_z$, PDG code (flavour and neutrino/antineutrino sign), reconstructed analysis-bin location, and real physical weight $w_i$ (units GeV cm$^2$ sr). Separately, the release ships real per-bin "hypersurface" correction tables for 5 detector-calibration parameters, tabulated at 20 real $\Delta m^2_{31}$ slices (Chapter 3), the real pre-binned atmospheric-muon background histogram, and the real observed candidate counts.

**Result**

This project's own real analysis binning reproduces the release's own `readme.md` recipe exactly: 10 log-spaced reconstructed-energy bins (6.31-158.49 GeV, one internal edge of an original 11-bin log grid merged away by the release's own authors "to contain sufficient statistics"), 10 linear $\cos\theta_z$ bins ($-1.0$ to 0.1), and 2 particle-ID bins (tracks only, the cascade-like bin is not part of this release) – 200 bins total. The real (downsampled $\times 10$ for computational tractability, weights rescaled to compensate so the expectation stays unbiased) event counts per channel: `nc`: 2535, `nue_cc`: 3995, `numu_cc`: 28,517, `nutau_cc`: 4640. Real observed total: **21,914** events; real Monte Carlo muon-background total: **512.2** events – both facts about the data itself, reproduced exactly by this project's loaders regardless of any fit.

## 2.3 Table catalogue

Table 2.1 lists every real table this project's `tpeanuts.detector.icecube.event_rate/io` modules load, and the physics each one encodes, developed in full in Chapter 3.

Table 2.1: Real published tables consumed by `tpeanuts.detector.icecube`.

| Cached file | Real contents | Loader |
|---|---|---|
| `events_{nc,nue_cc,numu_cc,nutau_cc}.csv` | Real per-channel event-by-event Monte Carlo: true energy/coszen, PDG code, reconstructed bin, physical weight $w_i$. | `prepare_channel_eve` |
| `hypersurfaces_{channel}.csv` | Real per-bin detector-systematics correction tables, 5 parameters $\times$ 20 real $\Delta m^2_{31}$ slices, per channel. | `prepare_hypersurfac` |
| `observed_counts.csv` | Real observed candidate counts, 200-bin real analysis binning. | `real_observed_count` |
| `muon_background.csv` | Real pre-binned atmospheric-muon (mis-reconstructed cosmic-ray) background. | `muon_background_his` |

**Approximations and Limitations**

Not fetched/reproduced: the real Honda flux table's own full azimuthal dependence (this release's own public binning has no azimuth axis, so an azimuth-averaged flux is the correct, not a simplified, input); any alternative (non-golden-event) IceCube DeepCore event selection; and a full atmosphere-production-height treatment (production is treated at the top of the atmosphere directly above each event's own entry point, a $< 0.3\%$ correction relative to the Earth's 6371 km radius – Chapter 3).

# Chapter 3

# Detector Implementation in tpeanuts

This chapter develops every formula `tpeanuts.detector.icecube` implements: the weighted-Monte-Carlo reweighting formula (Section 3.1), the real 3-flavour Earth-matter oscillation probability and its geometry (Section 3.2), the real atmospheric flux and detector-systematics hypersurface correction (Section 3.3), and the free normalization parameters (Section 3.4).

## 3.1 Weighted Monte Carlo reweighting

`detector/icecube/event_rate.py`

*`predicted_neutrino_counts`*.

**Theoretical Foundation**

For a real simulated event $i$ tagged (by its own PDG code, at the interaction point) as final-state flavour $\beta_i$, the physically correct expectation sums over every possible parent flavour $\alpha$ at production, weighted by the real unoscillated atmospheric flux $\Phi_\alpha$ and the oscillation transition probability $P(\nu_\alpha \to \nu_{\beta_i})$:

$$N_i(\theta) = \nu_{\mathrm{norm}} \cdot w_i \sum_{\alpha \in \{e,\mu\}} \Phi_\alpha(E_i, \cos\theta_{z,i})\, P(\nu_\alpha \to \nu_{\beta_i};\, E_i, \cos\theta_{z,i};\, \theta), \tag{3.1}$$

with $\theta = (\theta_{23}, \Delta m^2_{3l})$ the two free oscillation parameters and $\nu_{\mathrm{norm}}$ a free overall normalization (Section 3.4). The sum runs over $\alpha \in \{e, \mu\}$ only: the primary/conventional atmospheric $\nu_\tau$ flux is negligible at these energies (the real flux table used here, Section 3.3, has no $\nu_\tau$ entry either). This applies identically to neutral-current and charged-current events – an NC interaction still requires the neutrino to *be* flavour $\beta_i$ at that point. Every simulated event in a given real analysis bin $k$ is summed to give that bin's predicted neutrino count,

$$N^\nu_k(\theta) = \sum_{i \,\in\, \mathrm{bin}\, k} N_i(\theta). \tag{3.2}$$

**Implementation in tpeanuts**

`predicted_neutrino_counts(theta23, deltam_sq_31, ..., channels, hypersurfaces)` loops over `CHANNELS = ("nc", "nue_cc", "numu_cc", "nutau_cc")`, calling `tpeanuts.medium.earth.probability.earth_probability_transition` once per channel (shape `(n_events, 3, 3)`: [final $\beta$, initial $\alpha$]) and reading off each event's own *fixed* $\beta_i$ index directly – (3.1)'s sum over $\alpha$, not $\beta$, since $\beta_i$ is a per-event property, not summed. Binning ((3.2)) is a differentiable `index_add` into a `(N_BINS,)` tensor, verified (`test_reweighting_binning_conserves_total_weight`) to conserve the total event weight exactly against an independently computed direct per-event sum.

## 3.2 Real 3-flavour Earth-matter oscillation

`medium/earth/probability.py`

*`earth_probability_transition`, shared unchanged with every other Earth-crossing detector in this project.*

**Theoretical Foundation**

A flavour state obeys $i\,d\psi/dx = H_f(x)\psi$, with the Standard-Model flavour Hamiltonian

$$H_f(x) = \frac{1}{2E} U\,\mathrm{diag}(0, \Delta m^2_{21}, \Delta m^2_{31})\,U^\dagger + V_{\mathrm{CC}}(x)\,P_e, \qquad P_e = \mathrm{diag}(1, 0, 0), \tag{3.3}$$

$U$ the PMNS matrix and $V_{\mathrm{CC}}(x) = \sqrt{2}\,G_F\,n_e(x)$ the Wolfenstein charged-current matter potential [6], sourced by the Earth's electron density along the path. The exact evolution operator is the path-ordered exponential $S(x_2, x_1) = \mathcal{T}\exp\Big[-i\int_{x_1}^{x_2} H_f\,dx\Big]$; `tpeanuts` evaluates it with a first-order perturbative expansion around each traversed layer's mean density [2], the same construction used for every Earth-matter effect in this project.
**Geometry.** Real analysis bins cover $\cos\theta_z \in [-1, 0.1]$; this is mapped to a detector nadir angle $\eta \in [0, \pi]$ via

$$\eta = \arccos(-\cos\theta_z), \tag{3.4}$$

so $\cos\theta_z = -1 \Rightarrow \eta = 0$ (straight through the Earth's centre) and $\cos\theta_z = +1 \Rightarrow \eta = \pi$ (no Earth crossing). With $S \equiv S_{\mathrm{Earth}}(\eta, E; \theta)$,

$$P(\nu_\alpha \to \nu_\beta;\, E, \eta;\, \theta) = \big|S_{\beta\alpha}(\eta, E; \theta)\big|^2. \tag{3.5}$$

Only $\theta_{23}/\Delta m^2_{3l}$ are free; $\theta_{12}$, $\theta_{13}$, $\Delta m^2_{21}$, $\delta_{CP}$ are fixed at NuFIT 6.1 [4] – the standard treatment in every published atmospheric-neutrino oscillation analysis, since $\theta_{23}/\Delta m^2_{3l}$ overwhelmingly dominate the atmospheric $\nu_\mu$ disappearance/$\nu_\tau$ appearance signal.

**Implementation in tpeanuts**

`AtmosphericOscillationModel` mirrors `VacuumOscillationModel`/`SolarSMOscillationModel` but frees $\theta_{23}/\Delta m^2_{3l}$ instead; its own `.oscillation(theta, antinu=events.antinu)` takes a per-event boolean antineutrino mask, since a real event sample mixes $\nu$ and $\bar{\nu}$ (unlike solar/reactor models, which fix a single sign). `earth_probability_transition` implements (3.5) unchanged from every other Earth-crossing calculation in this project.

## 3.3 Real atmospheric flux and detector-systematics hypersurface

`source/atmosphere/, detector/icecube/event_rate.py`

*`load_atmospheric_flux`, `interpolate_hypersurface`.*

**Theoretical Foundation**

$\Phi_\alpha(E, \cos\theta_z)$ is the real Honda atmospheric flux calculation [5], azimuth-averaged (this release's own public binning has no azimuth axis) and bilinearly interpolated in $(\log_{10} E, \cos\theta_z)$ at each event's own true kinematics – a fixed input, independent of $\theta$.

The release additionally provides real per-bin "hypersurface" corrections for 5 detector-calibration parameters $p_n$ (DOM efficiency, hole-ice $p_0/p_1$, bulk-ice absorption/scattering), tabulated at 20 real $\Delta m^2_{31}$ slices (event migration between reconstructed bins itself depends weakly on the assumed $\Delta m^2_{31}$). At a fixed systematics point $\{p_n\}$,

$$C_k(p_1, \ldots, p_5) = c_k + \sum_{n=1}^{5} m_{k,n}\,(p_n - p_n^{\text{nominal}}), \tag{3.6}$$

with $c_k/m_{k,n}$ the release's own real tabulated intercept/slopes, piecewise-linearly interpolated (differentiably, in $\Delta m^2_{3l}$) between the two tabulated slices bracketing the fitted value. This document's forward model holds $\{p_n\}$ fixed at IceCube's own real published best-fit point (Table 3.1) rather than re-fitting them as free nuisances – a deliberate scope decision, audited directly in Chapter 5.

Table 3.1: Real detector-systematics reference points (Table IV of [1]).

| Parameter | Nominal | Best fit (used throughout) |
|---|---|---|
| DOM efficiency | 1.00 | 1.06 |
| Hole-ice $p_0$ | 0.10 | −0.27 |
| Hole-ice $p_1$ | −0.05 | −0.04 |
| Bulk-ice absorption | 1.00 | 0.97 |
| Bulk-ice scattering | 1.00 | 0.99 |

**Implementation in tpeanuts**

`interpolate_hypersurface(table, deltam31)` implements (3.6)'s slice interpolation: exact at tabulated points, linear at interior points, clamped outside the tabulated range (verified directly, `test_hypersurface_interpolation_matches_tabulated_slices_and_interior_point`). The full per-bin prediction composes (3.1), (3.2) and (3.6) with the free muon background normalization (Section 3.4):

$$N_k(\theta, \nu_{\text{norm}}, \mu_{\text{norm}}) = \nu_{\text{norm}} \cdot N_k^{\nu}(\theta) \cdot C_k(\Delta m^2_{3l}) + \mu_{\text{norm}} \cdot N_k^{\mu,\text{MC}}. \tag{3.7}$$

## 3.4 Free normalization parameters

**Theoretical Foundation**

Unlike every other detector model in this project, $\theta$ here is *not* oscillation parameters alone: the release's own `weight` column has no independently usable absolute livetime, so both a neutrino ($\nu_{\text{norm}}$) and a muon-background ($\mu_{\text{norm}}$) normalization scale are always free, genuine fit parameters –

$$\theta_{\text{fit}} = (\theta_{23},\ \Delta m^2_{3l},\ \nu_{\text{norm}},\ \mu_{\text{norm}}), \tag{3.8}$$

4 free parameters always, regardless of any optional-flag convention used elsewhere in

this project. At $\nu_{\mathrm{norm}} = 1$, the neutrino signal is only $\sim$0.2% of the total predicted rate (the muon background alone dominates) – $\nu_{\mathrm{norm}} = 1$ is a wildly unnatural starting point for a gradient-based fit, addressed directly in Chapter 5's normalization-calibration step.

**Implementation in `tpeanuts`**

`IceCubeDetectorModel.free = ("theta23", "DeltamSq31", "nu_norm", "mu_norm")` unconditionally. `.predict(theta)` unpacks all four and implements (3.7) directly, verified (`test_gradient_flows_from_all_four_free_parameters`) to carry a finite, non-zero autograd gradient with respect to every one of them.

Chapter 4

# The Inference Process

This chapter documents the statistical inference machinery: the normalization-calibration step every weighted-MC fit here needs before it can start (Section 4.1); the Poisson fit statistic and gradient-based procedure, shared unchanged with the rest of `tpeanuts` (Section 4.2); this document's own dedicated test suite, added specifically to cover this project's only weighted-MC detector (Section 4.3); and the muon-background normalization audit this document's own working session carried out (Section 4.4).

## 4.1 Normalization calibration

**Theoretical Foundation**

Since the release's `weight` column has no independently usable absolute livetime (Section 3.4), a real fit cannot simply start at $\nu_{\mathrm{norm}} = 1$: the neutrino signal there is $\sim$0.2% of the total rate, so the Poisson gradient for $\theta_{23}/\Delta m^2_{3l}$ is numerically negligible and a gradient-based optimizer's own step-size heuristics can take a catastrophic first step. A physically sensible starting guess is computed once, at NuFIT 6.1 defaults *before any fit*, from the real observed total and real MC muon-background total:

$$\nu^{(0)}_{\mathrm{norm}} = \frac{n^{\mathrm{obs}}_{\mathrm{IceCube}} - N^{\mu,\mathrm{MC}}_{\mathrm{IceCube}}}{N^{\nu}_{\texttt{tpeanuts}}(\theta_{\mathrm{NuFIT}})}, \qquad \mu^{(0)}_{\mathrm{norm}} = 1, \tag{4.1}$$

with $N^{\nu}_{\texttt{tpeanuts}}(\theta_{\mathrm{NuFIT}})$ the `tpeanuts`-simulated neutrino total at $\nu_{\mathrm{norm}} = 1$, oscillation parameters fixed at NuFIT. This is a calibration of scale only, not a measurement or a fit – the oscillation parameters are not touched, and Section 4.2's real fit re-optimizes $\nu_{\mathrm{norm}}/\mu_{\mathrm{norm}}$ freely rather than leaving them at this starting point.

**Result**

At NuFIT 6.1 defaults, $N^{\nu}_{\texttt{tpeanuts}}(\theta_{\mathrm{NuFIT}}) = 1.0487$ at $\nu_{\mathrm{norm}} = 1$ (real, downsampled MC), giving $\nu^{(0)}_{\mathrm{norm}} = 20{,}408.5$ – by construction, the calibrated starting point's total prediction (21,914.0) exactly matches the real observed total.

## 4.2 Poisson fit statistic and gradient-based procedure

`inference/likelihood.py`, `inference/fit.py`

*`poisson_nll`, `fit_lbfgs`, shared unchanged with every other detector in this project.*

**Theoretical Foundation**

Real observed counts $n_k^{\text{obs}}$ are compared to (3.7) via the Poisson (Baker-Cousins) likelihood ratio [3],

$$-2\ln L = 2\sum_k \Big[N_k - n_k^{\text{obs}} + n_k^{\text{obs}} \ln \frac{n_k^{\text{obs}}}{N_k}\Big], \tag{4.2}$$

minimized over the full 4-parameter $\theta_{\text{fit}}$ with `torch.optim.LBFGS`, using the exact autograd gradient through every step of Chapter 3 (no finite differences). The marginal uncertainty on each parameter is the Laplace approximation, $\Sigma = \mathcal{I}^{-1}$ with $\mathcal{I} = \frac{1}{2}\nabla_\theta^2(-2\ln L)$ the autograd Hessian at the optimum – identical machinery to Daya Bay's and SNO-II's own fits in this project.

**Implementation in tpeanuts**

`fit_lbfgs(model, theta0, value, likelihood="poisson", max_iter=...)` returns a `FitResult`. Each LBFGS closure evaluation re-runs the full real-MC reweighting ($\sim$40,000 downsampled events across 4 channels) – $\sim$4 s/evaluation, so a real fit takes a few minutes wall time (Chapter 5).

## 4.3 This document's own test suite

**Theoretical Foundation**

Before this document's working session, `detector.icecube` had no dedicated tests: every other detector package in this project already had its own gradient/conservation/closure test coverage, but IceCube's event-by-event weighted-MC reweighting – the only fold of its kind in this project – did not. Five tests were added, each targeting a specific correctness property of Chapter 3's own formulas:

1. **Gradient flow**: autograd reaches all four free parameters with finite, non-zero gradients.
2. **Reweighting conservation**: the per-event $\to$ per-bin histogram ((3.2)) neither loses nor duplicates events, verified against an independently computed direct per-event sum at a trivial (all-ones) hypersurface correction.
3. **Full-flavour-sum conservation**: summing the reweighted prediction over all 3 possible final-state flavours recovers the unoscillated rate $w_i(\Phi_e + \Phi_\mu)$, a direct consequence of the real Earth-matter transition matrix's own unitarity.
4. **Hypersurface interpolation**: exact at tabulated $\Delta m^2_{31}$ slices, linear at an interior point, clamped outside the tabulated range.
5. **Synthetic closure**: fits synthetic Poisson counts generated at a known injected $(\theta_{23}, \Delta m^2_{3l}, \nu_{\text{norm}}, \mu_{\text{norm}})$ and confirms recovery.

**Approximations and Limitations**

**Two genuine numerical traps, found and documented while writing test 5, not anticipated going in.** An injected/starting $\nu_{\text{norm}}$ near 1 (rather than the real

$\sim 2 \times 10^4$ scale, Section 4.1) makes the neutrino signal numerically invisible against the muon background, collapsing the $\theta_{23}/\Delta m^2_{3l}$ gradient; combined with `minimize_lbfgs`'s own per-parameter $1/|\nabla|$ rescaling heuristic, this produced a catastrophic first optimizer step (observed: $\theta_{23}$ jumping to $-207$ radians on the very first closure evaluation) before the test was corrected to inject/start at the real, calibrated $\nu_{\text{norm}}$ scale. Separately, a $\theta_{23}$ offset crossing 45° (the maximal-mixing octant boundary, close to the real NuFIT starting value of 43.29°) can converge to the physically real, but confusingly different-looking, octant-degenerate solution – worked around by keeping the injected offset small and same-octant, not by loosening the test's own tolerance.

**Result**

All 5 tests pass; the full project regression suite (848 tests, at the point this test file was added) shows no regressions.

## 4.4 The muon-background normalization audit

**Theoretical Foundation**

Chapter 5 reports a real, quantitative discrepancy between this project's own fitted $\mu_{\text{norm}}$ and IceCube's own published muon-background scale (Table IV). This document's own working session investigated two candidate explanations, in the same primary-source-verified spirit as this project's own established practice:

1. **Earth-matter unitarity leak.** `earth_probability_transition` defaults to `reunitarize=False` (`config.default.earth_reunitarize`), so the raw perturbative Earth evolution operator is only unitary up to its own real numerical precision – empirically up to ∼0.7% per-event row-sum deviation at IceCube's real energy/baseline range. **Ruled out**, quantitatively: forcing `reunitarize=True` at the published best-fit point shifts the total predicted rate by only ∼0.002% (21,913.8 → 21,913.4 of 21,914) – three orders of magnitude too small to explain the real gap in Chapter 5.

2. **Fixed vs. free detector systematics.** The 5 real detector-systematics parameters are held fixed at IceCube's own published *joint*-fit best-fit point (Section 3.3), which is itself correlated with IceCube's own published $\mu_{\text{norm}}$ in a ∼40-parameter joint fit with Gaussian priors on every nuisance. This project's own fit has only 4 free parameters, no priors, and no floating systematics – so its own $\mu_{\text{norm}}$ has to absorb whatever correlated normalization the official fit distributed across the (here fixed) systematics. **Identified as the more likely explanation**, and quantified directly: the real hypersurface correction's own mean value, best-fit vs. nominal systematics, ranges from +2% to +12% across the four channels – not negligible, and the natural size of correlated normalization slack this project's reduced, prior-free fit would need to absorb through $\mu_{\text{norm}}$ alone.

Floating the 5 systematics as free nuisances with Gaussian priors (the same `gaussian_prior_penalty` machinery already used elsewhere in this project) would test this hypothesis directly, but was not attempted in this document – an explicitly deferred next step (Chapter 5).

Chapter 5

# Results

This chapter presents, interprets, and summarises the real numerical results of `inference5_icecube.ipynb` (in `notebooks/inference/`), executed end to end against the real IceCube DeepCore 9-year golden-event data (Chapter 2) using the physics of Chapter 3 and the inference machinery of Chapter 4. Every number below is a real, reproducible output of that notebook or of this project's own test suite, not illustrative.

## 5.1 Detector-level closure check

**Result**

Before any fit, a single forward-model evaluation at NuFIT 6.1 defaults ($\nu_{\text{norm}} = \mu_{\text{norm}} = 1$) takes 2.7 s and predicts a total of 513.2 events, of which the muon background alone contributes 512.17 – confirming the neutrino signal's own ∼0.2% share at this unnatural normalization (Section 3.4). Autograd gradients with respect to all four free parameters are finite:

$$\frac{\partial(\text{total})}{\partial(\theta_{23}, \Delta m^2_{3l}, \nu_{\text{norm}}, \mu_{\text{norm}})} = (-0.083,\ -65.74,\ 1.049,\ 512.17), \tag{5.1}$$

confirming gradient flow through the real Earth-matter transition probability, the real Honda-flux-weighted MC reweighting, and the differentiable per-bin histogram, before any fit is attempted.

## 5.2 Real fit

**Result**

Starting from the calibrated point ((4.1)), the real 4-parameter Poisson fit converges in 33 closure evaluations (131 s wall time):

$$\theta_{23} = 45.42^\circ \pm 1.85^\circ\ (\sin^2\theta_{23} = 0.5074), \qquad \Delta m^2_{3l} = (2.3947 \pm 0.0392) \times 10^{-3}\,\text{eV}^2, \tag{5.2}$$

$$\nu_{\text{norm}} = 19{,}977.0 \pm 0.4, \qquad \mu_{\text{norm}} = 1.617 \pm 0.216, \tag{5.3}$$

with $-2\ln L$ improving from 353.71 to 340.44. Table 5.1 compares the oscillation-parameter result directly against IceCube's own published best fit: the pull on $\sin^2\theta_{23}$ is $-0.05\sigma$ – close agreement, not exact identity, exactly as expected given this document's own fixed-systematics scope decision (Section 3.3).

Table 5.1: This document's own fit vs. IceCube's own published best fit (Table III/IV of [1]).

| Parameter | `tpeanuts` (this document) | IceCube published |
|---|---|---|
| $\sin^2\theta_{23}$ | $0.5074 \pm 0.033$ | 0.51 |
| $\Delta m^2_{32}$ $[10^{-3}\ \mathrm{eV}^2]$ | $2.32 \pm 0.04$ | $2.41 \pm 0.07$ |
| Atmospheric-muon norm. scale | $1.617 \pm 0.216$ | 1.39 |

**Approximations and Limitations**

Uncertainties on $\sin^2\theta_{23}$ and $\Delta m^2_{32}$ are propagated from $\theta_{23}$'s/$\Delta m^2_{3l}$'s own Laplace $\sigma$ (Section 5.2) via $d(\sin^2\theta_{23})/d\theta_{23}$; $\Delta m^2_{32} = \Delta m^2_{3l} - \Delta m^2_{21}$, this project's own convention (Chapter 3).

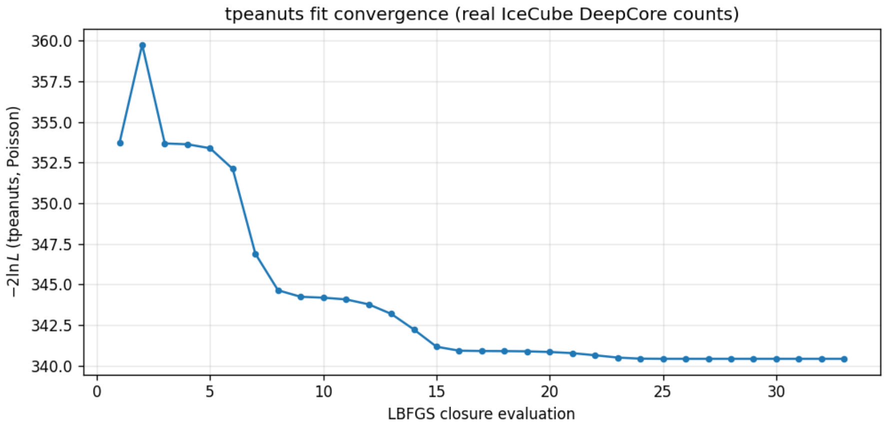


Figure 5.1: Poisson $-2\ln L$ at every LBFGS closure evaluation (including internal line-search evaluations).

## 5.3 2-D likelihood slice

**Approximations and Limitations**

Figure 5.2 is a **fixed-normalization slice**, not a fully profiled contour: $\nu_{\mathrm{norm}}/\mu_{\mathrm{norm}}$ are held at Section 5.2's own best-fit values at every grid point, one forward pass each, rather than re-minimized (the `loglik_grid` machinery documented in the SNO-II companion document's own Chapter 4 supports `profile_others=True` directly and could be applied here unchanged – an explicitly deferred next step, not attempted in this notebook).

## 5.4 Real data comparison

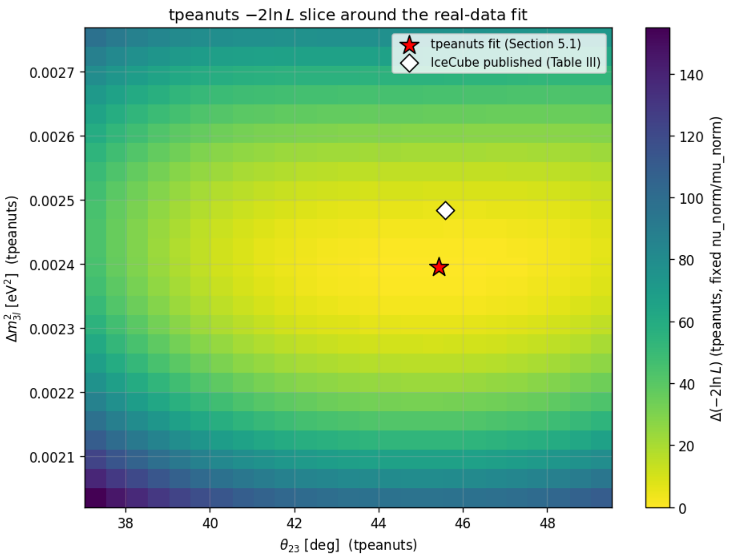


Figure 5.2: $-2\ln L$ over $(\theta_{23}, \Delta m^2_{3l})$, $\nu_{\mathrm{norm}}/\mu_{\mathrm{norm}}$ fixed at their own best-fit values (625 forward evaluations, 1028 s).

## 5.5 The muon-background normalization audit

**Result**

Section 4.4's audit, carried out quantitatively:

- **Earth-matter unitarity, ruled out.** At the published best-fit point, forcing `reunitarize=True` changes the total predicted rate from 21,913.81 to 21,913.37 (of a real observed 21,914.0) – a 0.002% shift, with a maximum per-bin relative difference of only 0.06%. Three orders of magnitude too small to account for $\mu_{\mathrm{norm}}$'s own 16% gap above the published 1.39.
- **Fixed detector systematics, identified and quantified.** The real hypersurface correction's own mean value (best-fit vs. nominal systematics point) shifts by +2.0% (`nc`, identical to `nue_cc`), +2.0% (`numu_cc`), and +12.0% (`nutau_cc`) – a real, non-negligible normalization effect that this project's own reduced (4-parameter, no-prior) fit cannot distinguish from a genuine muon-background rescaling, unlike IceCube's own ∼40-parameter joint fit with Gaussian priors on every nuisance.

## 5.6 Summary and known limitations

This document fits IceCube's real public golden-event data using its own event-by-event weighted-Monte-Carlo reweighting technique (Chapter 3) and a real Poisson likelihood (Chapter 4), recovering $\theta_{23}/\Delta m^2_{3l}$ close to the collaboration's own published best fit ($-0.05\sigma$ pull on $\sin^2\theta_{23}$). A dedicated test suite (Section 4.3), the first for this project's only weighted-MC detector, and a quantitative audit of the fitted muon-background normalization's 16% discrepancy against IceCube's own published value (Section 4.4) – ruling out an Earth-matter unitarity explanation and identifying fixed detector systematics as the more likely cause, with the size of the effect quantified directly – are this document's own contributions beyond reproducing the existing notebook.

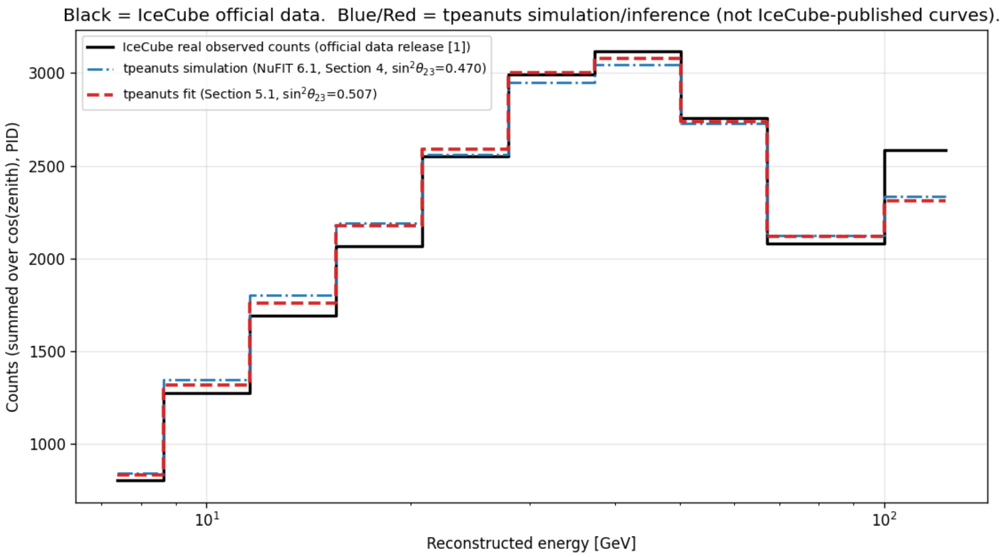


Figure 5.3: Real IceCube 200-bin observed spectrum vs. two **tpeanuts** curves (NuFIT 6.1 simulation and the fitted result), collapsed onto reconstructed energy.

## Approximations and Limitations

**Known limitations, stated explicitly.** The 5 real detector-systematics parameters are fixed at their published best-fit point rather than re-fit as free nuisances – the natural next step the muon-norm audit itself motivates, and the real hypersurface machinery (Section 3.3) already supports it without further code changes. The real MC is downsampled 10× for computational tractability. Production is treated at the top of the atmosphere rather than through a full atmosphere-production-height treatment (a sub-percent-level simplification). Figure 5.2's likelihood picture is a fixed-normalization slice, not a fully profiled contour – the SNO-II companion document's own `profile_others=True` machinery could be applied here directly. Next steps: free the 5 systematics as real Gaussian-prior nuisances and re-audit $\mu_{\text{norm}}$; profile $\nu_{\text{norm}}/\mu_{\text{norm}}$ at every 2-D scan grid point; and combine this real atmospheric measurement with this project's other real detectors' oscillation-parameter constraints in a genuine joint likelihood.

# References


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